

Error Detection & Recovery

Q. Explain the term error handling?

- o The process of locating errors and reporting them to users.

Q. What are the types of errors that occur commonly?

- o Globally there are two types of errors:

- Compile time errors
- Run time errors

Q. Enlist lexical Phase errors.

⇒ Typical lexical phase errors are

- o Spelling errors.

Hence get incorrect tokens. for example -

- o Exceeding length of identifiers
- o or numeric constants.
- o Appearance of illegal characters
- o Mistakes errors

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true examples of syntactical errors.

1. Syntax errors appear during syntax analysis phase of compiler.

2. Typical phase errors are:

- (1) Errors in structure misspelled keyword
 - (2) missing operators
 - (3) Unbalanced parenthesis e.g. ("hello";
 - (4) Extra blank space e.g. /* Command* /
 - (5) Colon in place of semicolon
- What are Semantic errors?

1. Semantic errors are those

errors which get detected during semantic analysis phase. Typical errors in this phase:

- (1) Incompatible types of operands e.g. $\begin{matrix} a & * & b \\ \text{int} & & \text{float} \end{matrix}$
- (2) undeclared variables
- (3) Not matching of actual parameters with formal arguments. e.g. $\text{int } a[10], b;$

a=b;

Q. Enlist various strategies to recover from syntactic errors.

⇒ Parser employs various strategies to recover from syntactic errors.

⇒ These strategies are:

- (1) Panic mode
- (2) Phrase level
- (3) Error productions
- (4) Global corrections

Q. What is panic mode strategy to recover from syntactic errors?

o In this method on discovering error, the parser discards input symbol one at a time.

This process is continued until one of a designated set of synchronizing tokens is found.

What is phase-level recovery?
In this method, on discovering errors passes perform local correction on remaining input.

It can replace a prefix of remaining input by some string. This actually helps the passes to continue its job.

The local correction can be replacing comma by semicolon, deletion of extra semicolons or inserting missing semicolon.

The type of local correction is decided by compiler designer.

This method is used in many error-reporting compilers.

Drawback:

- Difficult to handle the situations where the actual error has occurred before the point of detection.

Q. What is error production method in error recovery?

- o if we have a knowledge of common errors that can be encountered then we can incorporate these errors by augmenting the grammar of the corresponding language with error productions that generate the erroneous constructs.
- o if error production is used then during parsing, we can generate appropriate error message and parsing can be continued.
- o This method is difficult to maintain. Because if we change the grammar then it becomes necessary to change the corresponding error productions.

Q. What is global production in error recovery?

o A compiler requires that makes very few changes in processing an incorrect input string.

o We expect less number of insertions, deletions, and changes of tokens to recover from erroneous input.

o Such methods increase time and space requirements at parsing time.

o Global production is thus simply a theoretical concept.

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Exxox Recovery methods	Lexical Phase exxox	Syntactic Phase exxox	Semantic Phase exxox
Panic mode	✓	✓	X
Phrase level	X	✓	X
Exxox production	X	✓	X
Global production	X	✓	X
Using symbol table	X	X	✓